List of Symbols

M biomass.

S_i chemical species.

i index over chemical species.

j index over pathways.

i turnover rate.

1 Methods

1.1 Model

$$\frac{\mathrm{d}S_i}{\mathrm{d}t} = \sum_{j=1}^{N_P} M_j \left(v_j \cdot \gamma_{ij} - \delta\right) \tag{1}$$

$$v_j = v_{\max,j} \cdot f_{\mathrm{MM},j} \cdot f_{\mathrm{thermo},j} \cdot f_{\mathrm{inhib},j} \tag{2}$$

$$f_{\mathrm{MM},j} = \prod_{i=1}^{N_\mathrm{S}} \frac{S_i}{S_i + K_{m,ij}} \tag{3}$$

$$f_{\mathrm{inhib},j} = \prod_{i=1}^{N_\mathrm{S}} \frac{S_i}{S_i + K_{inhib,ij}} \tag{4}$$

If the i-th pool is irrelevant for the j-th pathway, $K_{m,ij} = 0$. Numerics: Add $\varepsilon = 10^{-12}$ in the denominator, to avoid division by zero.

$$H_{cc,i} = H_{cp}^{\circ} \cdot R \cdot T \cdot e^{C_{cp} \cdot \left(\frac{1}{T} - \frac{1}{T^\circ}\right)} \tag{5}$$

$$\lambda_i = H_{cc,i} \frac{V_\mathrm{aq}}{V_\mathrm{gas}} \tag{6}$$

$$S_i^\mathrm{dis} = \frac{\lambda_i}{1 + \lambda_i} \cdot S_i \tag{7}$$

$$a_i = \frac{S_i^\mathrm{dis}}{w} \tag{8}$$

$$Q_j = \prod_{i=1}^{N_P} a_i^{\gamma_{ij}} \tag{9}$$

$$\Delta G_{s,j} = \sum_{i=1}^{N_P} \gamma_{ij} \Delta G_{f,i} \tag{10}$$

$$\Delta G_{r,j} = \Delta G_{s,j} + RT \log Q_j \tag{11}$$

$$f_{\text{thermo},j} = 1 - e^{\frac{\min(0, \Delta G_{r,j} - \Delta G_{r,\min})}{RT}} \tag{12}$$

$\Delta G_{r,\min} = -26 \cdot 10^3 \text{-} \frac{\text{kJ}}{\text{mol}}$